BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: DIK-SSC-IX-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-IX-2022 (A)

Q1. Choose the correct option:

Q1. Which of the following units is a derived units?

(A) Kg
(B) Mole
(C) Cd
(D) N

Q2. 9.11 x 10^{-31}kg contains how many number of significant figures?

(A) 2
(B) 3
(C) 4
(D) 1

Q3. 295 km/h can be expressed in m/s as _____

(A) 81.9 m/s
(B) 8.19 m/s
(C) 819 m/s
(D) 819 km/s

Q4. A bird begins to accelerate at a constant 0.3m/s2 for 3s. Its change in velocity is _____

(A) 1.5 m/s
(B) 1.95 m/s
(C) 2.4 m/s
(D) 0.9 m/s

Q5. In 16N force is applied on a body of mass 4kg. How much acceleration will be produced in the body?

(A) 48 m/s^2
(B) 16 m/s^2
(C) 4 m/s^2
(D) 64 m/s^2

Q6. The unit of coefficient of kinetic friction is _____

(A) N
(B) μ
(C) μ_k
(D) No unit

Q7. Which of the following forces can act as a centripetal force?

(A) Tension
(B) Friction
(C) Gravitational force
(D) All of these

Q8. A paratrooper falling down is in dynamic translational equilibrium, its acceleration will be _____

(A) 9.8 ms^{-2}
(B) -9.8 ms^{-2}
(C) Zero
(D) Constant

Q9. The value of g_m on the surface of the moon is _____

(A) $g_e/6$
(B) $g_e/4$
(C) $g_e/3$
(D) $g_e = g_m$

Q10. Kilowatt hour is the unit of _____

(A) Power
(B) Energy
(C) Pressure
(D) Momentum

Q11. The unit used for pressure in weather map is _____

(A) Atm
(B) Pa
(C) Bar
(D) Nm^{-2}

Q12. When water at 0^0C is heated, it's volume decreases upto:

(A) 4^0C
(B) 10^0C
(C) 6^0C
(D) 100^0C

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 53 | **Paper Code:** DIK-SSC-IX-2022 (A)

Q2. Attempt any 8 questions of the following:

(i) Define least count of a screw gauge. How we can find least count of screw gauge?

(ii) A motorcyclist stats from rest and moves with uniform acceleration of $1.9m/s^2$ for 22s. Find: (a) Velocity and (b) Distance covered.

(iii) Can an object have a northward velocity and a southward acceleration? Explain.

(iv) Show that rate of change of momentum is equal to force.

(v) Why a balloon filled with air move forward, when its air is released?

(vi) What is neutral equilibrium? How an objects stability can be improved?

(vii) Define gravitational field strength. Gravity is a field force. How?

(viii) A 60g bullet is fired from a gun with 3150J of kinetic energy. Find its velocity.

(ix) Which material is more elastic, steel or rubber and why?

(x) Define specific heat capacity. What is the importance of high specific heat capacity of water?

(xi) An iron rim which is fixed around a wooden wheel is heated before its fixture. Explain why?

SECTION – C

Maximum Marks: 21

> Note: Attempt any Three Questions. All questions carry equal marks:

Q3.

(a) Graphically prove the third equation of motion i.e. $2as = vf^2 - vi^2$

(b) A golf bail that is initially traveling at 25m/s hits a sand trap and slows down with an acceleration of $20m/s^2$. Find its displacement after 2.0s.

Q4.

(a) Explain how value of g varies with attitude.

(b) Sana, who's mass is 43kg, sits 1.8m from the center of sea saw. Faiz, whose mass is 52kg, wants to balance Sana, How far from the center of the seesaw should Faiz sit?

Q5.

(a) State pascal principle, using pascal principle show that hydraulic lift is a force magnifier.

(b) When a weight of 30N is hung from a wire of original length 2m, its new length becomes 2.20m. Calculate the force constant for the wire, if the elastic limit is not exceeded?

Q6.

(a) Explain thermal conductivity and of a substance and its mathematical description.

(b) Find the amount of heat for melting the ice having mass 1.3kg at -10^0C. (Latent heat of fusion [illegible in source scan] $3.3x10^5J/kg$ and specific heat for ice $2.2x10^3J/kg$)